

Fig. 4: New Source Wizard window



Fig. 5: Adding ClockDivider.vhd in the design



Fig. 6: Skeleton source

ticked. If not, tick it.

5. Click Next and I/O port of the module will be displayed in the New Source Wizard window (Fig. 4).

6. Fill the name dcm in the entity field and enter CCLK_50MHz and RST as input ports and LOCK and CCLK_100MHz as output ports, and click Next and then the Finish button.

7. Select ClockDivider.vhd in the sources window and then click on New Source.

8. In New Source Wizard, select IP (Core Generator and Architecture Wizard) and enter DCM_Divide in the File name field and click Next.

9. In the Hierarchy window, ClockDivider.vhd is added as shown in Fig. 5 and a new skeleton source is created

with defined inputs and outputs as shown in Fig. 6.

10. The Select IP window appears on the New Source Wizard (Fig. 7). Click on View by Function. Select FPGA Features and Design → Clocking → Virtex-5. Your selection will depend on the FPGA series you are using.

11. Select Chose Wizard by Basic Function → Double Clock Frequency (DCM). Click Next and then Finish.

12. Xilinx Architecture Wizard-Setup window appears as shown in Fig. 8. Confirm that the Output File Type VHDL is selected, synthesis tool XST is selected and part number xc5vfx50-1ff676 is selected. (The part number should be the same as the part number entered by the designer earlier.)

13. Then the Xilinx Clocking Wizard—General Setup appears as shown in Fig. 9.

14. Enter 50 in the Input Clock Frequency field.

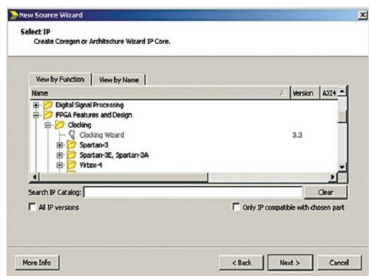


Fig. 7: Select IP window



Fig. 8: Xilinx architecture wizard-setup window

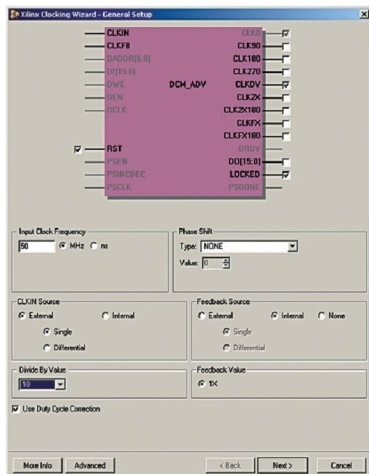


Fig. 9: Xilinx Clocking Wizard—General Setup window